Bayesian Statistics Project

Clustering Discrete Longitudinal Data with Mixtures of Generalized Linear Mixed Models

1 Objectives

The goal of this project is to design, implement, and validate a **clustering algorithm for non-Gaussian (discrete) longitudinal data**. Longitudinal (repeated-measures) studies frequently produce binary, count, or other non-Gaussian responses collected over time on each subject, and a natural scientific question is whether the subjects can be partitioned into a small number of homogeneous subgroups (clusters) according to the way their trajectories evolve. Because the response is discrete and observations within a subject are correlated, the problem cannot be handled by classical k-means or Gaussian mixture models directly, and it calls for methods built on *generalized linear mixed models* (GLMMs). You are free to design your own algorithm; the only requirements are that you read the three papers in Section 2, build your algorithm on their ideas, and extend them with at least one modern technique of your choice. All the questions you must answer are collected in Section 6.

2 Required Background Reading

Please read the following three papers carefully. Together they provide the model, the likelihood-based machinery, and the Bayesian machinery that your clustering algorithm should build upon.

1. **Zeger and Karim (1991)**, *Generalized Linear Models with Random Effects: A Gibbs Sampling Approach*, JASA 86, 79–86. This is the foundational paper for *fully Bayesian* inference in GLMMs. Study: (i) the Bayesian formulation (priors on fixed effects and on the variance components of the random effects); (ii) the derivation of the full conditional distributions; (iii) the use of the Gibbs sampler, including a Metropolis–Hastings step when a conditional distribution is not of a standard form; (iv) how posterior summaries and intervals are obtained from the MCMC output.

2. **McCulloch (1997)**, *Maximum Likelihood Algorithms for Generalized Linear Mixed Models*, JASA 92, 162–170. This paper reviews maximum-likelihood (rather than Bayesian) algorithms for GLMMs. Study in particular: (i) the basic model and notation (Section 2); (ii) the Monte Carlo EM (MCEM) algorithm of Section 3.1, in which the E-step is evaluated by Monte Carlo from the conditional distribution of the random effects given the data, combined with a Metropolis–Hastings step; (iii) the Monte Carlo Newton–Raphson (MCNR) algorithm of Section 3.2 and the simulated maximum likelihood (SML) method of Section 3.3; (iv) the practical issues (starting values, Monte Carlo sample size, convergence, and acceleration) discussed in Sections 4–7.

3. **Booth and Hobert (1999)**, *Maximizing Generalized Linear Mixed Model Likelihoods with an Automated Monte Carlo EM Algorithm*, JRSS-B 61, 265–285. This paper shows how to make MCEM *automated* and reliable. Study: (i) the two Monte Carlo approximations of the E-step — rejection sampling (Section 4.1) and importance sampling with a multivariate-t proposal (Section 4.2); (ii) the normal approximation

to the Monte Carlo error (Section 5); (iii) the stopping rules based on the Monte Carlo error (Section 6); (iv) how the Monte Carlo sample size is increased automatically as the algorithm converges.

3 Statistical Model and Notation

Consider n independent subjects. Subject i ($i = 1, \ldots, n$) is observed at T_i time points $t_{i1} < \cdots < t_{iT_i}$ and gives a *discrete* response Y_{ij} at each occasion ($j = 1, \ldots, T_i$). The responses within a subject are correlated, but subjects are independent of each other. We allow the number and spacing of time points to differ across subjects (this is more realistic than a balanced design; you may, of course, start with the balanced case $T_i \equiv T$ for convenience).

We assume that the n subjects can be partitioned into K latent clusters, where K may be fixed in advance or estimated from the data. Let $U_i \in \{1, \ldots, K\}$ denote the cluster membership of subject i, with

$$P(U_i = k) = \pi_k, \qquad k = 1, \ldots, K, \qquad \sum_{k=1}^{K} \pi_k = 1. \tag{1}$$

Conditional on $U_i = k$, the trajectory of subject i follows a GLMM with cluster-specific parameters. Specifically, let

$$g\big(\mu_{ijk}\big) = \eta_{ijk} = \mathbf{x}_{ij}^{\mathsf{T}} \boldsymbol{\beta}_k + \mathbf{z}_{ij}^{\mathsf{T}} \mathbf{b}_{ik}, \qquad \mu_{ijk} = \mathrm{E}\big[Y_{ij} \,\big|\, U_i = k, \mathbf{b}_{ik}\big], \tag{2}$$

where

- $g(\cdot)$ is a link function (e.g., the logit link for binary data, or the log link for counts);
- $\mathbf{x}_{ij}$ is a vector of fixed-effect covariates (typically including the time t_{ij} and, possibly, subject-level covariates), and $\boldsymbol{\beta}_k$ is the cluster-specific fixed-effect coefficient vector;

- $\mathbf{z}_{ij}$ is a sub-vector of $\mathbf{x}_{ij}$ (e.g., $z_{ij} = 1$ for a random intercept, or $z_{ij} = (1, t_{ij})$ for a random intercept plus random slope), and $\mathbf{b}_{ik}$ is the cluster-specific random effect of subject i, distributed as

$$\mathbf{b}_{ik} \sim N_q(\mathbf{0},\, \mathbf{D}_k), \tag{3}$$

  with $q = \dim(\mathbf{b}_{ik})$ and $\mathbf{D}_k$ a positive-definite covariance matrix.

The conditional distribution of Y_{ij} given $U_i = k$ and $\mathbf{b}_{ik}$ is a member of the exponential family with mean μ_{ijk}; e.g., Bernoulli with logit link, Poisson with log link, or negative binomial to accommodate overdispersion.

The unknown parameters are

$$\boldsymbol{\Omega} = \left\{\boldsymbol{\beta}_k,\, \mathbf{D}_k,\, \pi_k\right\}_{k=1}^{K}, \tag{4}$$

together with K itself if it is treated as unknown. Treating the cluster memberships U_i and the random effects $\mathbf{b}_{ik}$ as missing data, the complete-data likelihood is

$$L(\boldsymbol{\Omega} \mid \mathbf{Y}, \mathbf{U}, \mathbf{B}) = \prod_{i=1}^{n} \prod_{k=1}^{K} \Big\{ \pi_k\, \phi(\mathbf{b}_{ik}; \mathbf{0}, \mathbf{D}_k) \prod_{j=1}^{T_i} f_k(Y_{ij} \mid \mathbf{b}_{ik}) \Big\}^{\omega_{ik}}, \tag{5}$$

where $\phi(\cdot; \mathbf{0}, \mathbf{D}_k)$ is the $N_q(\mathbf{0}, \mathbf{D}_k)$ density, $f_k(Y_{ij} \mid \mathbf{b}_{ik})$ is the cluster-k conditional density of Y_{ij} (e.g., $\mu_{ijk}^{Y_{ij}}(1-\mu_{ijk})^{1-Y_{ij}}$ for binary data), and

$$\omega_{ik} = \begin{cases} 1, & \text{if subject } i \text{ belongs to cluster } k, \\ 0, & \text{otherwise.} \end{cases}$$

The observed-data likelihood,

$$L(\boldsymbol{\Omega} \mid \mathbf{Y}) = \prod_{i=1}^{n} \sum_{k=1}^{K} \pi_k \int \prod_{j=1}^{T_i} f_k(Y_{ij} \mid \mathbf{b}_{ik})\, \phi(\mathbf{b}_{ik}; \mathbf{0}, \mathbf{D}_k)\, \mathrm{d}\mathbf{b}_{ik}, \tag{6}$$

involves an intractable integral over the random effects, which is exactly the difficulty addressed by the three papers.

4 Methodological Approach

You are expected to *design your own algorithm.* The only requirements are that your algorithm (i) builds on the ideas and techniques of the three papers in Section 2, and (ii) incorporates at least one *modern* extension of your own choosing — for example variational Bayes, expectation propagation, or a Bayesian machine-learning method (such as stochastic variational inference, Hamiltonian Monte Carlo, or a Dirichlet-process mixture in which the number of clusters is inferred from the data). Apart from these constraints, you are free to decide the number of clusters, the estimation or inference strategy, the priors, and the details of the algorithm. A natural but by no means required strategy is to combine a likelihood-based EM-type method with a fully Bayesian approach and to compare them; the choice is yours, and part of the project is to identify and apply the methods yourself.

5 Simulation Study

Design a simulation study to (i) demonstrate that your algorithm recovers the true cluster structure, and (ii) quantify its accuracy and robustness. Report every detail so that the study is fully reproducible.

5.1 Data-generating process

Generate data from model (2)–(3) with known parameters. A sensible baseline is:

- $n = 300$ subjects and $T_i = T = 20$ common time points (balanced case), with a sensitivity check using unbalanced T_i and irregular time points;
- $K = 3$ clusters with mixing proportions $\boldsymbol{\pi} = (0.5, 0.3, 0.2)$;
- a random intercept ($z_{ij} = 1$) with $\mathbf{D}_k = \sigma_k^2$, e.g., $(\sigma_1^2, \sigma_2^2, \sigma_3^2) = (0.5, 0.5, 0.5)$;

- a binary response with logit link, $\eta_{ijk} = \beta_{0k} + \beta_{1k}\, t_{ij} + b_{ik}$, with clearly separated trajectories, e.g., $(\beta_{01}, \beta_{11}) = (-1.5, 0.2)$, $(\beta_{02}, \beta_{12}) = (0.5, -0.2)$, $(\beta_{03}, \beta_{13}) = (1.0, 0.3)$, over $t \in [-1, 1]$;
- additionally, a *count* response (Poisson log-linear, and negative binomial with overdispersion) to show that your method handles non-Gaussian discrete data beyond the binary case.

These values are suggestions; you are free (and encouraged) to choose your own, but you must state them explicitly.

5.2 Scenarios

Explore at least the following factors, one at a time, to assess robustness:

1. *Cluster separation*: vary the distance among $\boldsymbol{\beta}_k$ (easy vs. hard clustering problems).
2. *Sample size and series length*: vary n (e.g., 100, 300, 1000) and T (e.g., 10, 20, 50).
3. *Number of clusters*: fix K correctly, mis-specify K (too small or too large), and — if you implement a model-selection or nonparametric route — let K be chosen by the data.
4. *Variance components*: vary σ_k^2 to change the within-cluster heterogeneity.
5. *Response type*: binary vs. Poisson vs. overdispersed count.

6 Questions and Report Requirements

Write a *formal scientific report* (in Chinese or English) of no more than 15 pages (excluding figures and references), structured as a research paper: Abstract, Introduction (motivation and related work, citing the three papers), Model and Notation, Methods (deriva-

tions), Simulation Study (setup, results, tables and figures), Discussion, and References. In your report, answer *all* of the following questions.

(1) *Model and likelihood.* Give the model, the complete-data and observed-data likelihoods, and the form of the objective your algorithm actually evaluates (e.g., the Monte Carlo average of the log-likelihood, the evidence lower bound, or the posterior distribution).

(2) *Algorithm.* Write down, in full detail, the algorithm you implement: the E-step and M-step (for EM-type methods) or the full conditional distributions and update equations (for Gibbs/variational/EP methods), including every Metropolis–Hastings step with its proposal distribution and acceptance probability.

(3) *Initialization.* State your initial values, and explain how you avoid local optima and sensitivity to the starting point.

(4) *Convergence.* State your convergence criterion, and describe how you *assess the convergence* of your algorithm — e.g., the relative change of the objective, the Monte Carlo error, the evidence lower bound, trace plots, the $\widehat{R}$ statistic, or checks with multiple starting points.

(5) *Connection to the literature.* Explain how your algorithm draws on each of the three papers, and describe any acceleration or automation you adopted and the improvement you observed.

(6) *Modern extension.* Describe your modern extension (variational Bayes, expectation propagation, or a Bayesian machine-learning method), how it differs from and improves upon the classical method, and report timing and accuracy comparisons.

(7) *Simulation and clustering evaluation.* Present the simulation results using clustering evaluation criteria *of your own choosing* (for example, confusion matrix, misclassification rate, adjusted Rand index, or normalized mutual information), together with

the quality of the parameter estimation, and discuss how the results vary with n, T, cluster separation, the number of clusters, and the response type. Justify your choice of criteria.

(8) *Advantages and limitations.* Discuss the advantages and limitations of your algorithm. Be specific: when does it work well and when does it fail? Consider identifiability (label-switching), sensitivity to the choice of K, to initialization, and to the prior; computational cost and scalability to large n; the normality assumption on the random effects and the independence assumption across time; and the accuracy of the variational/EP approximation relative to exact MCMC.

(9) *Issues in the simulation design.* Discuss one or two issues you found in the simulation design and how you resolved them.

7 Assessment

Your project will be assessed on (i) correctness and depth of the statistical derivation; (ii) correctness of the implementation and the reproducibility of the simulation; (iii) the appropriateness and justification of the clustering evaluation; (iv) the depth and honesty of the discussion of advantages and limitations; and (v) the clarity and professionalism of the report.

References

[1] Zeger, S. L. and Karim, M. R. (1991). Generalized linear models with random effects: a Gibbs sampling approach. *Journal of the American Statistical Association*, 86, 79–86.

[2] McCulloch, C. E. (1997). Maximum likelihood algorithms for generalized linear mixed models. *Journal of the American Statistical Association*, 92, 162–170.

[3] Booth, J. G. and Hobert, J. P. (1999). Maximizing generalized linear mixed model likelihoods with an automated Monte Carlo EM algorithm. *Journal of the Royal Statistical Society, Series B*, 61, 265–285.